

Multiple Choice (10 marks)

1. A principal source of the Earth's internal energy is
 - (a) electromagnetic radiation.
 - (b) thermal conduction.
 - (c) gravitational pressure.
 - (d) radioactivity.
 - (e) nuclear fusion.
2. The negative muon, μ^- , *has properties most similar to which of the following?*
 3. Gluon
 4. Electron
 5. Quark
 6. Proton
 7. Photon

How many nuclei are there in 1 kg of aluminum?

- (a) 7.21×10^{25} nuclei
- (b) 9.09×10^{-26} nuclei
- (b) 4.48×10^{-26} nuclei
- (b) 26.98×10^{-27} nuclei
- (b) 3.01×10^{22} nuclei

The force required to start a mass of 50 kilograms moving over a rough surface is 343 Nt. What is the coefficient of starting friction?

- (a) 0.75
- (b) 0.40
- (c) 0.55
- (d) 0.65
- (e) 0.50

How are planetary rings made?

- (a) From accretion within the solar nebula at the same time the planets formed
- (b) From fragments of planets ejected by impacts
- (c) From dust grains that escape from passing comets
- (d) From the remnants of a planet's atmosphere
- (e) From the gravitational pull of nearby planets

True / False (10 marks)

Answer True or False

6. True or False: The centripetal force needed for the car to turn is significantly less than its weight, approximately 400 lb.
7. True or False: A minimum voltage of 65000 volts is sufficient to produce K radiation from a tungsten target in an x-ray tube.
8. True or False: The angular acceleration of the turntable, when accelerated from rest to 33.3 rpm in 2 seconds, is 1.74 sec^{-2} .
8. True or False: If the absolute temperature of a blackbody is increased by a factor of 3, the energy radiated per second per unit area increases by a factor of 81.
9. True or False: The magnetic field lines around a current-carrying wire form concentric circles that are oriented perpendicular to the wire.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the relationship between voltage and energy dissipation in a resistor. What happens to the rate of energy dissipation if the voltage is doubled? Provide a detailed analysis.
12. Calculate the time period of each oscillation of an ultrasonic transducer operating at 6.7 MHz and discuss how to derive its angular frequency in rad/s.

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